



Te Pūkenga

- School of Innovation, Design & Technology -

**IT5506 Bachelor of Information Technology
IT5487 Diploma in Information Systems
(Level 5)**

Practical Lab 1.2

Configure router and switch-based Ethernet network

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Score: / 100 points

Details

Due Date: **06 October 2022**

This Lab is worth **10%** of the over-all course grade

Overview

Learning Outcomes:

On successful completion of this assessment, the learner will be able to:

1. Describe network protocol models and devices to explain the layers of communications in data networks.
2. Design and calculate IP addresses and subnet masks for both IPv4 and IPv6 for given simple networks, using IPv4 and IPv6.
3. Explain fundamental Ethernet concepts.

Tasks

There are four parts to this assessment.

Part one you will use the Cisco Packet Tracer application to build a virtual network consisting of routers, switches, servers, printers and computers. The network will be built based on the network topology diagram using the appropriate network media to connect devices together.

Part two you will design an IPv4 network addressing scheme to use on your network devices. You will need to break the overall IPv4 network domain into smaller subnets. You will be using the Variable Length Subnet Mask (VLSM) subnetting technique to create these smaller broadcast domains. Once your IPv4 addressing scheme has been identified you will create a matching IPv6 hexadecimal addressing scheme ensuring devices have the same subnet id and host id of IPv4 and IPv6 addresses.

You will use the following addresses for your network.

- **192.168.10.0 /24**
- **2023:5506:5487:: /64**

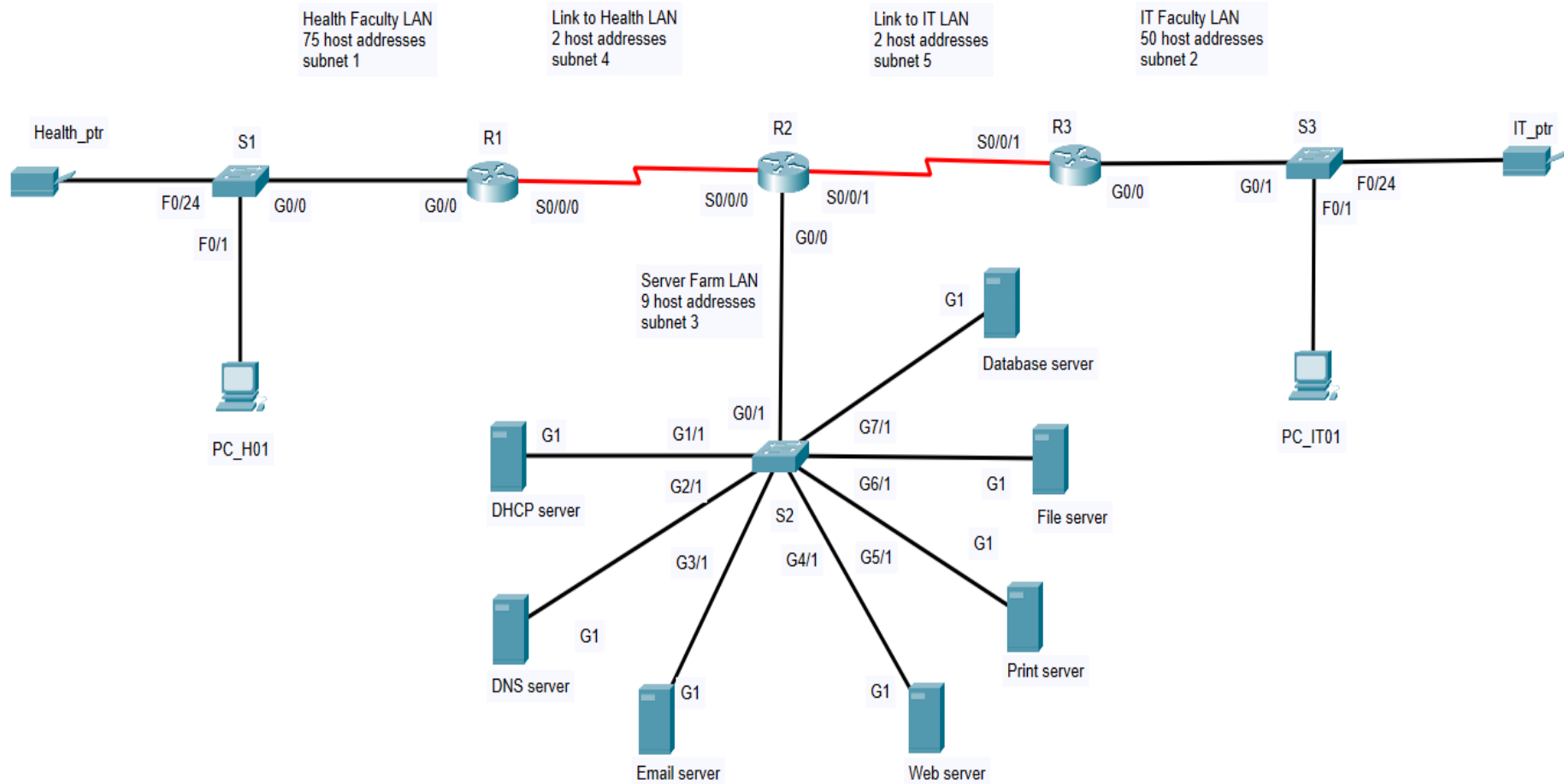
Part three you will configure network devices with the appropriate IPv4 and IPv6 addresses as identified in your IP Addressing Scheme. Network routers and switches also need basic security and routing configuration settings.

Part four you must verify connectivity between the computers and all other devices. All devices must be reachable via their IPv4 and IPv6 GUA addresses while switches will only be reachable via their IPv4 addresses.

There are two assessment files that you need to upload our Moodle page. These are to be completed and uploaded to Moodle Lab 1.2 Submission Box for marking.

1. Lab 1.2 PT assessment file
2. Lab 1.2 assessment tasks (this file)

Network Logical Topology Diagram



Part One: Build the Network

22 marks

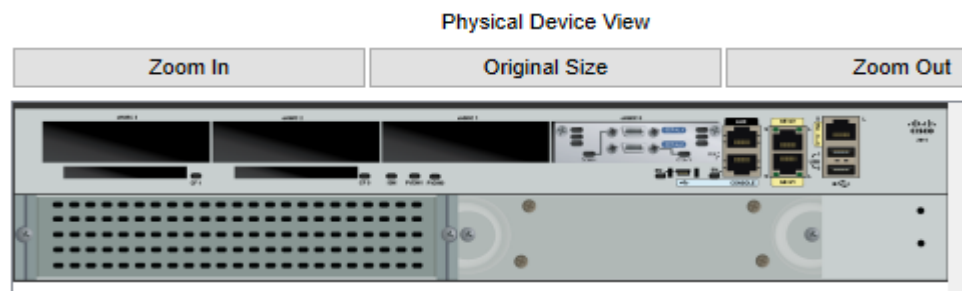
Using the Packet Tracer application build the virtual network based on the Network Topology Diagram above.

1. Open Packet Tracer application and create a new file.
2. Add all network devices
3. Add new interfaces and ports to routers, switches and servers
4. Connect media to appropriate interfaces and ports and NIC's
5. Add the following Routers to the desktop.
 - a. Click on the Network Devices icon
 - b. Click on the Routers icon
 - c. Select the 2911 Router and add 3 routers to the desktop
 - i. You will need to add serial interfaces to each router
 - ii. Click on R1 router and the Physical tab
 - iii. Turn the Power Off
 - iv. Click on the Modules tab to view interface modules to add
 - v. Click on the HWIC-2T module and drag it to the empty interface port closet to the existing LAN ports
 - vi. Power On the router
 - vii. Repeat these steps for R2 and R3 routers

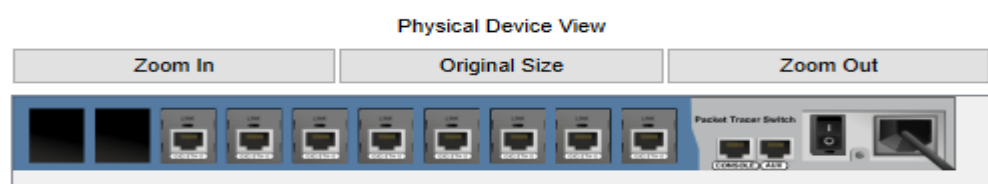
5 marks

9 marks

8 marks



6. Add the following Switches to the desktop
 - a. Click on the Network Devices icon
 - b. Click on the Switches icon
 - c. Select the 2960 Switch and add 2 switches to the desktop
 - d. These will be used for the IT and Health LAN's.
 - e. Now select the PT-Empty switch and add 1 switch to the desktop
 - i. This switch will be used for the Server Farm LAN
 - ii. You will need to add Gigabit Ethernet interfaces to the switch
 - iii. Click on the PT-Empty switch and the Physical tab
 - iv. Turn the Power Off
 - v. Click on the Modules tab to view interface modules to add
 - vi. Click on the PT-SWITCH-NM-1CGE module and drag it to the empty interface port closet to the existing Power button
 - vii. Repeat this step until you have added 8 new interfaces
 - viii. Power On the switch



7. Add the following End-user devices to the desktop eg computers, printers and servers
- Click on the End Devices icon
 - Click on the Printer icon and add 2 printers to the desktop
 - Click on the PC icon and add 2 PC to the desktop
 - Click on the Server icon and add 7 servers
 - Each server will need to have a Gigabit Ethernet NIC installed
 - Click on a server and the Physical tab
 - Turn the Power Off
 - Click on the Modules tab to view interface modules to add
 - Click on the PT-HOST-NM-1CGE module, drag to the empty NIC port
 - Turn the Power On
 - Repeat these steps for each server



8. Connect all network devices based on the network topology diagram
- Connect router to router
 - Click on the Connections icon
 - Click on the Serial DCE cable
 - Connect to R2 router serial interface
 - Connect to R1 router serial interface
 - Click on the Serial DTE cable
 - Connect to R2 router serial interface
 - Connect to R3 router serial interface
 - Connect routers to LAN switches
 - Click on the Connections icon
 - Select the appropriate utp patch cable to use
 - Connect to R1 router gigabit ethernet interface
 - Connect to S1 switch gigabit ethernet port
 - Click on the Connections icon
 - Select the appropriate utp patch cable to use
 - Connect to R2 router gigabit ethernet interface
 - Connect to S1 switch gigabit ethernet port
 - Click on the Connections icon
 - Select the appropriate utp patch cable to use
 - Connect to R3 router gigabit ethernet interface
 - Connect to S3 switch gigabit ethernet port
 - Connect Printers to LAN switches
 - Click on the Connections icon

- ii. Select the appropriate utp patch cable to use
 - iii. Connect to S1 switch fast ethernet port
 - iv. Connect to printer fast ethernet port
 - v. Select the appropriate utp patch cable to use
 - vi. Connect to S3 switch fast ethernet port
 - vii. Connect to printer fast ethernet port
- d. Connect Computers to LAN switches
 - i. Click on the Connections icon
 - ii. Select the appropriate utp patch cable to use
 - iii. Connect to S1 switch fast ethernet port
 - iv. Connect to computer fast ethernet port
 - v. Select the appropriate utp patch cable to use
 - vi. Connect to S3 switch fast ethernet port
 - vii. Connect to printer fast ethernet port
- e. Connect servers to LAN switch
 - i. Click on the Connections icon
 - ii. Select the appropriate utp patch cable to use
 - iii. Connect to DHCP server to S2 switch G1/1 gigabit ethernet port
 - iv. Connect to DNS server to S2 switch G2/1 gigabit ethernet port
 - v. Connect to Email server to S2 switch G3/1 gigabit ethernet port
 - vi. Connect to Web server to S2 switch G4/1 gigabit ethernet port
 - vii. Connect to Print server to S2 switch G5/1 gigabit ethernet port
 - viii. Connect to File server to S2 switch G6/1 gigabit ethernet port
 - ix. Connect to Database server to S2 switch G7/1 gigabit ethernet port

Part Two: Design the IP Addressing Scheme

36 marks

1. Network IP Address Requirements

You have been assigned the following network IP addresses to use. Based on these addresses you will create a subnet addressing scheme that will minimise the number of wasted IPv4 addresses.

IPv4 address	192.168.10.0 /24
IPv6 address	2023:5506:5487:: /64

Use the Variable-Length Subnet Mask (VLSM) subnetting method to subnet the existing IPv4 network address to meet the following network requirements. These are the minimum IP addresses needed for each LAN.

Note when using VLSM start from the subnet needing the most addresses first, then the next highest etc until you cover all subnet requirements

1. Health Faculty LAN needs 75 host addresses
2. IT Faculty LAN needs 50 host addresses
3. Server Farm LAN needs 9 host addresses
4. Link to IT LAN needs 2 host addresses
5. Link to Health LAN needs 2 host addresses

2. Complete the Binary IP Addressing table

In the following subnet addressing table using binary number system identify each subnets network address, first useable host address, last useable host address and the subnets broadcast address.

Subnetting questions.

5 marks

Identify number of host bits needed for each subnet and the total number of useable addresses available.

- a. Based on the original network address and subnet mask address (/24), how many host bits, and how many useable addresses in the original broadcast domain

256 host, 254 useable addresses,

- b. Health Faculty is the first subnet needed

128 host, 126 useable addresses, 255.255.255.128

- c. IT Faculty subnet is the second subnet needed

64 host, 62 useable addresses, 255.255.255.192

- d. Server Farm subnet is the third subnet needed

16 host, 14 useable addresses, 255.255.255.224

- e. Links to Health and IT LAN are the fourth and fifth subnets needed

4 host, 2 usable addresses, 255.255.255.252

3. Binary IPv4 Subnet Addressing Table

10 marks

Now complete the Binary IPv4 Addressing table to document the range of IPv4 addresses to use in each subnet and the decimal equivalent of the binary addresses.

Note: You only need to show the binary for the last 8 bits of the address eg for the address of 192.168.10.1 it would be shown as **00000001**. For the subnet mask write this down in the shortened slash format eg **/24**.

a. Subnet 1. Health Faculty	Binary IPv4 address	Decimal
Subnet network address	192.168.10.0 /25	00000000
First useable host address	192.168.10.1 /25	00000001
Last useable host address	192.168.10.126 /25	01111110
Subnet broadcast address	192.168.10.127 /25	01111111

b. Subnet 2 IT Faculty	Binary IPv4 address	Decimal
Subnet network address	192.168.10.128 /26	10000000
First useable host address	192.168.10.129 /26	10000001
Last useable host address	192.168.10.190 /26	10111110
Subnet broadcast address	192.168.10.191 /26	10111111

c. Subnet 3 Server Farm	Binary IPv4 address	Decimal
Subnet network address	192.168.10.192 /27	11000000
First useable host address	192.168.10.193 /27	11000001
Last useable host address	192.168.10.222 /27	11001110
Subnet broadcast address	192.168.10.223 /27	11001111

d. Subnet 4 and 5 LAN Links	Binary IPv4 address	Decimal
Subnet network address	192.168.10.224 /30	11100000
First useable host address	192.168.10.225 /30	11100001
Last useable host address	192.168.10.228 /30	11101110
Subnet broadcast address	192.168.10.229 /30	11101111

e. Reserved Addresses	Binary IPv4 address	Decimal
Subnet network address	192.168.10.230 /28	11110000
First useable host address	192.168.10.231 /28	11110001
Last useable host address	192.168.10.245 /28	11110110
Subnet broadcast address	192.168.10.246 /28	11110111

4. IP Address Allocation Specifications

Based on your IP addressing scheme use the appropriate subnet and host addresses as specified below. Then complete the IP Address Allocation Table.

Health Faculty Subnet

- R1 router G0/0 interface gets the first ip address of the subnet
- S1 switch vlan 1 interface gets the second ip address of the subnet
- P1 printer gets the last ip address of the subnet
- Host computers get rest of the subnet addresses starting from the third ip address of the subnet

IT Faculty Subnet

- R3 router G0/0 interface gets the first ip address of the subnet
- S3 switch vlan 1 interface gets the second ip address of the subnet
- P3 printer gets the last ip address of the subnet
- Host computers get rest of the subnet addresses starting from the third ip address of the subnet

Server Farm Subnet

- R2 router G0/1 interface gets the first ip address of the subnet
- S2 switch vlan 1 interface gets the second ip address of the subnet
- Server computers get rest of the subnet addresses starting from the third ip address of the subnet
 - DHCP server gets the third ip address of the subnet
 - DNS server gets the fourth ip address of the subnet
 - Email server gets the fifth ip address of the subnet
 - Web server gets the sixth ip address of the subnet
 - Print server gets the seventh ip address of the subnet
 - File server gets the eighth ip address of the subnet
 - Database server gets the ninth ip address of the subnet

LAN Link Subnets

1. R2 router S0/0/0 interface gets the first ip address of the subnet
2. R1 router S0/0/0 interface gets the second ip address of the subnet
3. R2 router S0/0/1 interface gets the first ip address of the subnet
4. R3 router S0/0/1 interface gets the second ip address of the subnet

Unused IP addresses

- The rest of the unused ip addresses are reserved for future use.

Once you have identified the IPv4 subnet addressing scheme you will then implement the IPv6 hexadecimal addressing scheme to match the IPv4 subnet and host addresses allocated to the different LAN's and complete the IP Address Allocation table below

5. IP Address Allocation Table

21 marks

Note IPv6 GUA addresses must match the IPv4 subnet id and host numbers but in hexadecimal format

Note IPv6 LLA addresses for the computers, printers and servers are automatically generated, do not change.

Note IPv4 you only need to show the last decimal digit of the address e.g. 192.168.10.1 would be .1 and /24 for the subnet mask

Device	Interface	IPv4	IPv4 DFG	IPv6 LLA	IPv6 GUA	IPv6 DFG
R1	G0/0	192.168.10.1	n/a	FE80::1:1	2023:5506:5487:1::1	n/a
	S0/0/0	192.168.10.226	n/a	FE80::4:2	2023:5506:5487:4::E2	n/a
R2	G0/0	192.168.10.193	n/a	FE80::3:1	2023:5506:5487:3::C1	n/a
	S0/0/0	192.168.10.225	n/a	FE80::4:1	2023:5506:5487:4::E1	n/a
	S0/0/1	192.168.10.229	n/a	FE80::5:1	2023:5506:5487:5::E5	n/a
R3	G0/0	192.168.10.129	n/a	FE80::2:1	2023:5506:5487:2::81	n/a
	S0/0/1	192.168.10.230	n/a	FE80::5:2	2023:5506:5487:5::E6	n/a
Switches						
S1	Vlan 1	192.168.10.2	192.168.10.1	n/a	n/a	n/a
S2	Vlan 1	192.168.10.194	192.168.10.193	n/a	n/a	n/a
S3	Vlan 1	192.168.10.130	192.168.10.129	n/a	n/a	n/a
Computers, Printers and Servers						
Health_Faculty_PC	NIC	192.168.10.3	192.168.10.1	n/a	2023:5506:5487:1::3	FE80::1:1
Health_Printer	NIC	192.168.10.126	192.168.10.1	n/a	2023:5506:5487:1::7E	FE80::1:1
IT_Faculty_PC	NIC	192.168.10.131	192.168.10.129	n/a	2023:5506:5487:2::83	FE80::2:1
IT_Printer	NIC	192.168.10.190	192.168.10.129	n/a	2023:5506:5487:2::BE	FE80::2:1
DHCP	NIC	192.168.10.195	192.168.10.193	n/a	2023:5506:5487:3::C3	FE80::3:1
DNS	NIC	192.168.10.196	192.168.10.193	n/a	2023:5506:5487:3::C4	FE80::3:1
Email	NIC	192.168.10.197	192.168.10.193	n/a	2023:5506:5487:3::C5	FE80::3:1
Web	NIC	192.168.10.198	192.168.10.193	n/a	2023:5506:5487:3::C6	FE80::3:1
Print	NIC	192.168.10.199	192.168.10.193	n/a	2023:5506:5487:3::C7	FE80::3:1
File	NIC	192.168.10.200	192.168.10.193	n/a	2023:5506:5487:3::C8	FE80::3:1

Database	NIC	192.168.10.201	192.168.10.193	n/a	2023:5506:5487:3::C9	FE80::3:1
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Part Three: Configure the Network Devices

32 marks

Now that the design phase is complete start configuring the devices. Based on your addressing table you can configure IP address settings for all devices.

1. Computer configuration

2 marks

- Configure the ipv4 address, subnet mask DNS server, and ipv4 default gateway
- Configure ipv6 GUA address, prefix, DNS server and ipv6 default gateway
- Accept the default LLA ipv6 address

2. Printer configuration

2 marks

- Assign an appropriate display name to the printers e.g. IT_ptr and Health_ptr
- Configure the ipv4 address, subnet mask, DNS server and ipv4 default gateway
- Configure the ipv6 GUA address, prefix, DNS server and ipv4 default gateway

3. Server configuration

7 marks

- Assign an appropriate display name to servers eg. DHCP, DNS
- Configure the ipv4 address, subnet mask DNS server, and ipv4 default gateway
- Configure ipv6 GUA address, prefix, DNS server and ipv6 default gateway
- Accept the default LLA ipv6 address

4. Switch configuration

9 marks

- For each switch configure the following settings

Mark	Configure these settings
1	Set Hostname to S1, S2 or S3
1	Set the secret password to: itsasecret
1	Set console password to: letmein
1	Set vty access password to: class
1	Configure a login message of the day to: "Authorized Access only!"
1	Set the IPv4 default gateway address based on addressing tables
1	Configure VLAN 1 interface address based on addressing tables
1	Encrypt all passwords
1	Save the configuration to NVRAM

5. Router configuration

12 marks

- For each router configure the following settings

Mark	Configure these settings
1	Enable IPv6 unicast-routing Note: This MUST BE THE FIRST command inputted to enable IPv6 routing
1	Set Hostname to R1, R2 or R3
1	Set the secret password to: itsasecret
1	Set console password to: letmein
1	Set vty access password to: class
1	Configure a login message of the day to: "Authorized Access only!"
3	Configure serial and ethernet interfaces based on addressing tables description

	• ipv4 address and subnet mask • ipv6 GUA address and prefix • ipv6 LLA address • activate the interface
2	Configure IPv4 and IPv6 static routes to other LAN networks
1	Save the configuration to NVRAM

Part Four: Connectivity Testing

10 marks

Time to conduct connectivity testing. You will test ipv4 and ipv6 GUA address connectivity.

1. Using the ping command conduct connectivity testing from the IT computer to all other devices. If any of the connectivity tests fail troubleshoot the problems and fix any connectivity issues.
2. **Ping connectivity test results table**

Device	IPv4 address	IPv6 GUA address
IT_printer	192.168.10.190	2023:5506:5487:2::BE
S1 VLAN 1 interface	192.168.10.2	n/a
R1 G0/0 interface	192.168.10.1	2023:5506:5487:1::1
R1 S0/0/0 interface	192.168.10.226	2023:5506:5487:4::E2
R2 S0/0/0 interface	192.168.10.225	2023:5506:5487:4::E1
R2 S0/0/1 interface	192.168.10.229	2023:5506:5487:5::E5
R3 S0/0/1 interface	192.168.10.230	2023:5506:5487:5::E6
R3 G0/0 interface	192.168.10.129	2023:5506:5487:2::81
S3 VLAN 1 interface	192.168.10.130	n/a
Health_printer	192.168.10.126	2023:5506:5487:1::7E
Health computer	192.168.10.3	2023:5506:5487:1::3
R2 G0/1 interface	192.168.10.193	2023:5506:5487:3::C1
S2 VLAN interface	192.168.10.194	n/a
DHCP server	192.168.10.195	2023:5506:5487:3::C3
DNS server	192.168.10.196	2023:5506:5487:3::C4
Email server	192.168.10.197	2023:5506:5487:3::C5
Web server	192.168.10.198	2023:5506:5487:3::C6
Print server	192.168.10.199	2023:5506:5487:3::C7
File server	192.168.10.200	2023:5506:5487:3::C8
Database server	192.168.10.201	2023:5506:5487:3::C9

Assessment complete.

Upload this file and your Packet Tracer file to Moodle

Marking Schedule

Part One: Build the Network	22 Marks
Add all devices 1 mark for each set of devices added e.g., 3 routers, 3 switches, 7 servers, 2 printers and 2 computers	5 marks
Update interfaces on routers, switches and servers 0.5 marks for each interface added 3 router serial interfaces 8 switch Gigabit Ethernet ports 7 server Gigabit Ethernet NIC's	9 marks
Connect devices using ports specified in the Topology Diagram 0.25 for each correct cable to port connections 2 serial cables 14 ethernet utp cables	8 marks
Part Two: Design the IP Addressing Scheme	36 Marks
2a – 2e 1 mark per question	5 marks
3a – 3e Binary IPv4 Subnet Addressing Tables 2 marks per table	10 marks
5. IP Address Allocation Table 1 mark for each line of interface address settings	21 marks
Part Three: Configure the Network Devices	32 Marks
1 configure computers	2 marks
2 configure printers	2 marks
3 configure servers	7 marks
4 configure switches	8 marks
5 configure routers	13 marks
Note to marker • Mark both computers. 1 mark per correct ipv4 and ipv6 addresses. Add total and divide by 8. Use this average mark as the result • Mark both printers. 1 mark per correct ipv4 and ipv6 address. Add total and divide by 8. Use this average mark as the result • Mark all 7 servers. 1 mark per correct ipv4 and ipv6 address. Add total and divide by 8. Use this average mark as the result • Mark all 3 switches add totals and divide by 3 to get the average mark for configuration tasks. Use this average mark as the result • Mark all 3 routers add totals and divide by 3 to get the average mark for configuration tasks. Use this average mark as the result	
Part Four: Connectivity Testing	10 Marks
2. Ping connectivity test results table 0.25 marks per address successfully tested	10 marks
Total	100 Marks